

Byzantine Robustness of Federated Learning for IoT Anomaly Detection

Data-Poisoning Attack Simulation on the N-BaIoT Dataset

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1 Context and Motivation

Federated Learning (FL) distributes training across client devices without centralising raw data. This architecture, while privacy-preserving, exposes the global model to *Byzantine* clients — participants that submit corrupted model updates to degrade the shared model. In the IoT threat-detection context studied here, a Byzantine client corresponds to a compromised device that deliberately trains on a mixture of normal and abnormal traffic, forcing the global model to treat attack signatures as normal behaviour.

This report quantifies how the two FL models from the main comparison study — **Autoencoder (AE)** and **Hybrid Shrink-Autoencoder (SAE)** — respond when an increasing fraction of the 10-client IID federation is corrupted.

2 Experimental Setup

2.1 Attack Model

The attack implemented is **data poisoning**: a Byzantine client replaces 30 % of its normal training samples with randomly drawn abnormal traffic samples before local training. The poisoned client still participates normally in FedAvg aggregation; the corrupted gradients propagate into the global model without any detection mechanism. No model-weight manipulation is applied — the corruption is entirely at the data level.

2.2 Protocol

All hyperparameters are kept identical to the clean FL baseline (Table 1), so that Byzantine performance can be directly compared against clean performance.

Table 1: Fixed hyperparameters shared with the clean FL baseline.

Parameter	Value
Dataset distribution	IID, 10 clients
FL rounds	20 (with early stopping, patience = 2)
Epochs per round	100
Client participation	50 % per round (5 of 10)
Independent runs	5
Learning rate	10^{-5}
Shrink λ	10
Latent dimension	11
Poisoning ratio	30 % of selected Byzantine client’s training set
Byzantine ratios swept	10 %, 20 %, 30 %, 40 %, 50 %

2.3 Evaluation Metric

At the end of each FL round, the global model is evaluated on every client’s held-out test set (normal + abnormal traffic). The **best-round ROC-AUC** — the round yielding the highest mean AUC across all 10 clients — is reported per run, then averaged across the 5 independent runs. This mirrors the metric used in the clean FL comparison paper.

3 Results

3.1 Summary: Mean AUC vs. Byzantine Ratio

Table 2 reports the best-round mean AUC $\pm$ standard deviation across the 5 runs for each (model, Byzantine ratio) combination. The clean FL baseline (0 % Byzantine, from the main comparison study) is included as the reference row.

Table 2: Best-round mean AUC (%) across 10 IID clients and 5 runs. Degradation Δ is relative to the clean baseline.

Byzantine %	Autoencoder		Hybrid (SAE)	
	Mean	Std	Mean	Std
0 % (clean)	99.07	— — —	99.01	— — —
10 %	99.09	0.06	98.11	0.31
20 %	99.12	0.03	98.41	0.40
30 %	99.09	0.06	97.58	0.87
40 %	99.10	0.06	97.59	0.83
50 %	99.05	0.06	96.54	0.66
Δ (0% $\rightarrow$ 50%)	−0.02		−2.47	

Key observation. The Autoencoder is virtually immune to data-poisoning attacks at the best-round level: AUC fluctuates by less than 0.10 percentage points across all Byzantine ratios (99.05–99.12 %). The Hybrid model degrades monotonically from 98.11 % at 10 % corruption to 96.54 % at 50 % corruption — a −2.47 pp drop relative to the clean baseline of 99.01 %. Variance also increases markedly for the Hybrid (std = 0.87 at 30 %), indicating that the interaction between Byzantine clients and the centroid-based one-class classifier is sensitive to which specific clients are selected as adversaries.

3.2 AUC Degradation Over FL Rounds

While the best-round AUC metric captures peak performance, it does not reveal how quickly the global model degrades as poisoned updates accumulate. Tables 3 and 4 report the mean AUC across all 10 clients at each FL round (averaged over the 5 runs). Rows are truncated at the point where early stopping ends the run.

Table 3: Autoencoder — mean AUC (%) per FL round by Byzantine ratio (IID, mean of 5 runs).

Byz%	Round																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
10	99.09	98.99	98.51	98.01	97.89	97.38	96.69	95.66	94.74	94.90	94.32	94.33	93.08	93.17	93.84	93.77	93.90	94.35	93.67	93.74
20	99.12	99.07	98.41	97.51	97.20	97.17	96.63	95.96	95.65	95.19	95.31	94.49	94.47	94.61	94.35	94.69	94.70	94.92	94.41	94.68
30	99.08	99.02	98.51	98.63	98.58	98.30	97.89	97.53	97.37	96.44	95.92	96.01	95.74	95.73	95.09	95.64	94.97	95.01	94.85	94.78
40	99.09	99.05	98.75	98.64	98.49	98.34	97.82	97.25	97.20	96.29	96.17	95.61	95.99	96.06	95.86	95.85	95.47	95.51	95.23	95.15
50	99.05	99.00	98.71	98.38	98.17	98.09	98.02	97.81	97.35	97.50	97.39	96.89	97.01	96.36	96.86	96.49	96.55	96.42	96.04	96.11

Table 4: Hybrid (SAE) — mean AUC (%) per FL round by Byzantine ratio (IID, mean of 5 runs).

Byz%	Round																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
10	96.96	96.55	97.40	96.91	97.32	97.64	97.29	97.35	96.95	97.35	97.40	97.35	96.99	97.21	97.27	97.33	97.24	97.24	97.34	97.34
20	96.78	96.51	96.99	96.73	96.92	97.67	96.87	96.69	97.10	96.61	97.56	96.80	97.39	97.05	97.05	97.55	97.12	97.54	97.35	97.19
30	96.55	96.17	96.40	95.87	96.19	96.59	96.23	96.24	96.20	96.16	96.02	95.97	96.30	96.26	96.08	96.23	96.50	—	—	—
40	95.56	96.18	96.22	95.36	96.48	96.71	96.55	96.36	96.98	96.90	96.93	96.98	96.74	96.85	96.83	96.97	96.88	96.90	96.98	97.00
50	94.54	95.19	94.55	95.05	95.57	95.95	95.77	96.25	—	—	—	—	—	—	—	—	—	—	—	—

Autoencoder round dynamics. All ratios start at round-1 AUC of $\approx 99.08\%$ and decline monotonically as poisoned updates accumulate. Higher Byzantine ratios slow the decline: at byz10%, AUC falls to 93.08% by round 13; at byz50%, it stays above 96.04% throughout 20 rounds. This counter-intuitive pattern reflects the stochastic client selection (50% per round): with more Byzantine clients in the pool, any given round has a higher probability of selecting none or few of them, so the corrupted updates are diluted by more honest rounds. Regardless, the best-round metric picks round 1 in all cases, confirming that the optimal AE model under Byzantine attack is the one obtained after a single FL round.

Hybrid round dynamics. The Hybrid model converges differently: round-1 AUC starts significantly lower ($\approx 94\text{--}97\%$) and generally improves or stabilises across rounds as the centroid in the latent space stabilises. At byz50%, early stopping at round 8 prevents further training; at byz10% and byz20%, the model trains the full 20 rounds with stable performance. This confirms that the centroid-based classifier benefits from more FL rounds to define a reliable normal-traffic cluster — but Byzantine poisoning compresses the improvement window by destabilising the latent representation.

3.3 Per-Client AUC at Maximum Corruption (50%)

Table 5 breaks down AUC by client at the worst-case scenario (byz50%). Values are the mean across the 5 runs at the best round.

Table 5: Per-client best-round AUC (%) at byz50% (mean of 5 runs, IID).

Client	Autoencoder	Hybrid (SAE)
Client-1	99.50	93.19
Client-2	99.66	99.63
Client-3	95.62	94.49
Client-4	97.67	90.94
Client-5	99.73	99.14
Client-6	99.40	94.69
Client-7	99.78	96.92
Client-8	99.52	97.17
Client-9	99.85	99.42
Client-10	99.81	99.82
Mean	99.05	96.54

Two clients stand out persistently across all experiments:

- **Client-3** has the lowest AUC for the Autoencoder (95.62%) regardless of Byzantine ratio. This is an inherent property of its data distribution in the IID split, not an effect of the attack.
- **Client-4** achieves 97.67% under AE but collapses to 90.94% under the Hybrid at byz50%. The Hybrid’s centroid-based classifier is more sensitive to distributional shifts in the latent space; when Byzantine clients poison the latent representation, the centroid shifts away from Client-4’s normal-traffic cluster.

4 Discussion

4.1 Why the Autoencoder Is Robust

The Autoencoder detects anomalies by reconstruction error: test samples with high MSE are flagged as abnormal. Data poisoning introduces abnormal samples into the training set, which teaches the model to reconstruct attack patterns. However, because only 30 % of a Byzantine client’s training data is abnormal, and Byzantine clients represent at most 50 % of participants, the effective fraction of poisoned samples across the federation in any given round is at most 15 % (0.5×0.3). The aggregated global model remains dominated by the honest majority, and its reconstruction surface for normal traffic changes only slightly. This explains why round-1 AUC — before repeated poisoning can shift the surface — is essentially identical to the clean baseline.

The decline over subsequent rounds is real but mitigated by FedAvg’s weight averaging: each Byzantine client contributes a weight proportional to its dataset size, which is not inflated in this attack model (data poisoning does not increase the client’s apparent dataset size).

4.2 Why the Hybrid Model Is More Vulnerable

The Hybrid adds a centroid-based one-class classifier in the latent space on top of the reconstruction model. Its anomaly score is a combination of reconstruction error *and* distance from the centroid of normal traffic in the latent space. This second signal is sensitive to the quality of the latent representation: when Byzantine clients inject abnormal samples, the encoder is subtly trained to compress abnormal patterns into the latent space, shifting the centroid towards attack traffic. Even a small shift degrades the distance-based classifier for honest clients.

Variance in the Hybrid results (std up to 0.87 %) reflects this sensitivity: the attack impact depends on which specific clients are selected as Byzantine each run, since different clients have different latent representations of normal traffic.

4.3 Practical Implication

Under data-poisoning attacks with a 30 % poisoning ratio and up to 50 % Byzantine clients in a 10-device IID federation:

- The **Autoencoder** maintains ≥ 99.05 % AUC and is the recommended architecture when Byzantine robustness is a design requirement.
- The **Hybrid model** provides superior AUC on clean data (99.01 %) and its convergence over rounds is more stable, but it is susceptible to degradation under heavy attack (96.54 % at 50 % Byzantine). Deploying the Hybrid requires either a Byzantine-robust aggregation rule (e.g., Krum, coordinate-wise median) or a trust mechanism to limit participation of suspected compromised clients.
- **Early stopping** is insufficient as the sole defence: the validation loss can keep decreasing even as the anomaly-detection AUC degrades (the model becomes better at reconstructing poisoned data, so MSE on the poisoned validation set drops). Round-level AUC monitoring on a small held-out reference set is a more reliable stopping criterion.

5 Conclusion

This study simulated Byzantine data-poisoning attacks on a 10-client IID Federated Learning system trained for IoT anomaly detection. Five Byzantine ratios (10–50 %) were evaluated across two model architectures and 5 independent runs.

The **Autoencoder** is highly resilient: best-round AUC degrades by only 0.02 pp from the clean baseline across all attack intensities (99.05–99.12 %). Its robustness stems from FedAvg’s weight averaging, which dilutes poisoned updates, and from the reconstruction-error mechanism, which is less sensitive to distributional shifts in the latent space.

The **Hybrid Shrink-Autoencoder** shows a clear degradation trend, losing 2.47 pp at 50 % corruption (96.54 % vs. 99.01 % clean). The centroid-based classifier is the vulnerable component, as it depends on a clean latent representation of normal traffic.

These findings motivate follow-up work on Byzantine-robust aggregation (Krum, coordinate-wise median) and Byzantine-tolerant centroid estimation, which could recover the Hybrid model’s clean-data advantage while maintaining robustness under attack.

A Appendix: Per-Client AUC by Ratio

Tables 6 and 7 give the full per-client AUC breakdown for all Byzantine ratios (mean of 5 runs).

Table 6: Autoencoder — per-client best-round AUC (%) by Byzantine ratio.

Client	10 %	20 %	30 %	40 %	50 %
Client-1	99.53	99.56	99.54	99.52	99.50
Client-2	99.66	99.66	99.66	99.68	99.66
Client-3	95.57	95.62	95.62	95.62	95.62
Client-4	97.71	97.86	97.71	97.74	97.67
Client-5	99.77	99.78	99.76	99.77	99.73
Client-6	99.56	99.58	99.53	99.55	99.40
Client-7	99.83	99.83	99.83	99.83	99.78
Client-8	99.55	99.61	99.56	99.58	99.52
Client-9	99.88	99.88	99.88	99.88	99.85
Client-10	99.80	99.84	99.80	99.82	99.81
Mean	99.09	99.12	99.09	99.10	99.05

Table 7: Hybrid (SAE) — per-client best-round AUC (%) by Byzantine ratio.

Client	10 %	20 %	30 %	40 %	50 %
Client-1	98.54	98.80	96.88	96.76	93.19
Client-2	99.64	99.61	99.63	99.59	99.63
Client-3	94.90	94.67	94.13	94.14	94.49
Client-4	95.95	96.61	95.00	94.85	90.94
Client-5	99.14	99.38	99.13	99.00	99.14
Client-6	97.60	98.13	96.55	97.07	94.69
Client-7	97.69	98.60	97.31	96.96	96.92
Client-8	98.76	98.77	98.20	98.35	97.17
Client-9	99.12	99.70	99.20	99.45	99.42
Client-10	99.80	99.79	99.78	99.78	99.82
Mean	98.11	98.41	97.58	97.59	96.54